

MICHELE DE CILLIS

Data Engineer

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EXPERIENCE

Research Intern

NERDS Lab @ ITU

📅 Sep 2025 – Dec 2025 📍 Copenhagen, DK

Conducted bachelor's thesis research on measuring polarization in large-scale directed social networks during an Erasmus traineeship, including a literature review across social media analysis, network polarization, and network science. Stack: Python, NumPy, PyTorch, Pandas.

- Re-engineered an HPC data pipeline processing the **entire historical Reddit dataset**, converting it to build directed instead of undirected networks
- Measured polarization on directed networks by applying the Magnetic Laplacian — the **first use in this context**
- Validated results against theoretical toy-network models, reaching a **0.65 average correlation**
- Earned the **highest thesis grade (7/7)**. Read [here](#)

Co-founder

Polyneta

📅 Apr 2025 – Dec 2025 📍 Remote

Co-founded a startup connecting companies that sell to municipalities to simplify direct-award procurement, running customer discovery and building the early product with AI-accelerated development.

- Prototyped an ETL pipeline (scraping, RAG, and OCR) to extract **structured data** from disorganized public-sector PDFs
- Built the alpha (pre-MVP) aggregating company services for municipalities, using AI to accelerate development

Front-End Engineer

Pane&Design (PED, part of humanvsmachine.co)

📅 Oct 2021 – Dec 2022 📍 Milan, IT

Built and optimized high-traffic e-commerce sites and internal telemedicine tools for a Fortune 500 client at a digital consulting agency, using React, TypeScript, styled-components, Tailwind, and Material UI.

- Cut re-renders and load times by **perceived 50%** on a high-traffic e-commerce site (**100k monthly visits**) via microservices with dynamic JS loading
- Built a telemedicine tool for the client's ophthalmologists, unifying a multi-section medical history form into a single-page flow with react-jsonschema-form

SKILLS

Language and Tools - Python

PostgreSQL Docker Nix Bash

Linux Git TypeScript JavaScript

GraphQL HTML/CSS Figma Typst

LaTeX

Frameworks and Libraries - NumPy

Pandas PyTorch DuckDB

scikit-learn Matplotlib Node.js

React React Native TanStack Query

LANGUAGES

Italian Native speaker

English C1 Advanced

French B1

Spanish B1

EDUCATION

BSc Computer Science

Università degli Studi di Milano

📅 Sep 2022 – Apr 2026

Calculus Linear Algebra Physics

Data Structures and Algorithms

Erasmus+ Exchange Program

Université Paris-Saclay

📅 Sep 2023 – Jun 2024

Bio Informatics Machine Learning & NLP

Linear Programming

CERTIFICATIONS

Cambridge C1 Advanced • 📅 26 Mar 2026

Full-Stack Engineer

Online Impresa (part of areaweb.org)

📅 Jul 2020 – Sep 2021 📍 Bari, IT

Sole developer at a small software agency serving SMEs, building the company's flagship mobile app platform from scratch with TypeScript, React Native, GraphQL, and Expo, and working directly with clients on requirements.

- Built an automated build pipeline that clones the template repo, loads client specs, and builds with Expo, cutting deployment **from 1 hour to 15 minutes**
- Built internal reporting tools, including a live office dashboard showing the **daily count of closed orders**, fed by a WhatsApp bot posting every closed order
- Delivered **100 client apps**, gathering and implementing requirements directly with clients

PROJECTS

ALS Biomarker Identification

Analyzed RNA-Seq data and developed a model that generalizes with 90% accuracy for the ALS group. Authored comprehensive documentation of a workflow to identify potential biomarkers.

📅 May 2024 – May 2024

Personal Homelab

Self-host personal services (Nextcloud, Immich, and more) on a VPS with NixOS, using reproducible configuration-as-code to add services and their infrastructure (backups, nginx) quickly, with the same config shared across devices. Uses Tailscale for VPN, nginx as a reverse proxy, and daily BorgBase backups.

📅 Mar 2025 – Present

Open Source Contributions

📅 2025 – Present

- Ported and maintaining a package in [nixpkgs](#) (+25k stars, Nix language)
- Usability bugfixes (mainly state sync) in [Pixelfed](#) (+7k stars, React Native)
- Improved the Italian translation in [Ghost Blog](#) (+54k stars)

ACCOMPLISHMENTS

Top-Percentile Academic Ranking

University of Milan

📅 2024/25

Ranked in the top 20% for academic performance and the top 15% for graduation grade within the reference academic year.

First Place

University Startup Challenge

📅 2025

Won first place as a team of 4 by conducting market research, gathering early feedback, and building the pitch, landing page, and Figma prototype for what became Polyneta.

Silicon Valley Study Tour

📅 Aug 2025

Selected for a competitive program visiting leading companies and universities (Cisco, Apple, Google, LinkedIn, Electronic Arts, PureStorage, Sysdig, Zoox, Stanford, Berkeley), attending keynotes from their teams and building connections.